

Stakeholder Engagement Plan: Agriculture and Climate Domain-Specific SLM

1. Purpose

The development of an Agriculture and Climate Small Language Model can affect multiple groups of people. Stakeholder engagement is therefore important to ensure that the system is relevant, responsible, understandable, and aligned with real-world needs.

This engagement plan identifies key stakeholders and describes how their perspectives can contribute to the development and improvement of the project.

2. Key Stakeholders

Farmers and Agricultural Producers

Farmers are potential end users of agriculture-related AI systems.

Their feedback is important for understanding:

- common agricultural questions;
- local farming practices;
- information needs;
- practical usability of generated answers;
- local terminology and context.

Future engagement could include interviews, surveys, demonstrations, or feedback sessions.

Agricultural Experts and Agronomists

Agronomists and agricultural specialists can help assess the quality and safety of generated information.

Their role may include:

- reviewing agricultural recommendations;
- identifying factual errors;
- validating domain-specific knowledge;
- suggesting additional training data.

Expert feedback is particularly important before using the system in real-world advisory settings.

Climate and Environmental Experts

Climate specialists can contribute expertise related to:

- climate adaptation;
- changing environmental conditions;
- weather and climate risks;
- sustainable agricultural practices.

Their involvement can help improve the scientific relevance of climate-related information.

Researchers and AI Practitioners

Researchers and AI practitioners contribute to:

- model design;
- dataset development;
- training methods;
- evaluation;
- responsible AI practices.

They can also help identify technical limitations and improve reproducibility.

Local Communities

Agricultural knowledge is often connected to local environments and practices.

Community engagement can help ensure that future versions of the system respect:

- local knowledge;
- cultural context;
- regional agricultural practices;
- local information needs.

Policy Makers and Development Organizations

Government institutions, NGOs, and development organizations may have an interest in AI applications for agriculture and climate resilience.

Their perspectives can help identify potential areas where responsible AI tools could support existing agricultural and climate initiatives.

3. Engagement Approach

The project follows a progressive engagement approach.

Stage 1: Understanding Needs

Identify common agriculture and climate information needs through discussions with potential users and domain experts.

Stage 2: Dataset and Knowledge Review

Invite domain experts to review the relevance, quality, and limitations of the knowledge used to train the system.

Stage 3: Prototype Demonstration

Demonstrate the prototype to selected stakeholders and collect feedback about:

- answer relevance;
- clarity;
- usefulness;
- usability;
- potential risks.

Stage 4: Iterative Improvement

Use stakeholder feedback to improve:

- dataset quality;
- prompt design;
- model behavior;
- evaluation methods;
- safety measures.

4. Feedback Mechanisms

Possible feedback mechanisms include:

- surveys;
- interviews;
- focus group discussions;
- prototype demonstrations;
- expert reviews;
- structured evaluation forms.

Feedback should focus on whether generated answers are understandable, relevant, contextually appropriate, and safe.

5. Responsible Engagement

Stakeholder engagement should follow responsible principles.

Participants should understand the purpose of the project and how their feedback may be used.

The project should avoid collecting unnecessary personal or sensitive information.

Stakeholder feedback should be treated as valuable input for improving the system, particularly when identifying limitations or potential harms.

6. Future Engagement

This initial project provides a prototype for further development.

Future work should prioritize stronger collaboration with:

- farmers;
- agricultural extension services;
- agronomists;
- climate experts;
- local communities.

Such collaboration can help ensure that future versions of the Agriculture and Climate SLM are more accurate, locally relevant, and practically useful.